



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

SECR1033 - 09 COMPUTER ORGANIZATION AND ARCHITECTURE

GROUP PROJECT - CPU & Hardware Performance Benchmarks

Lecturer: Dr Nur Haliza binti Abdul Wahab

Name	Matric No
Loh Hui Yi	A23CS0106
Lee Jian Ai	A23CS0234

Video Link: [YouTube](#)

Table of Content

1.0 Introduction.....	2
2.0 Part 1.....	5
2.1 Overview.....	5
2.2 The Flowchart of the Project.....	6
2.3 Benchmarking Results.....	7
2.4 Analysis, Comparison, Discussion.....	14
3.0 Part 2.....	23
3.1 Overview.....	23
3.2 The Flowchart of the Execution.....	24
3.3 Coding and Documentation.....	25
3.4 Implementation / Execution and Results (print screen outputs).....	33
3.5 Analysis, Comparison, Discussion.....	39
4.0 Conclusion and Reflection.....	42
References.....	44
Appendices.....	45

1.0 Introduction

Benchmarking is the most frequent method for obtaining information on processor and computer system performance, and it is used to compare one system's performance to another objectively. CPU benchmarking is a series of tests meant to assess the performance of a computer or device's CPU. A collection of standards or basic metrics is used to compare the performance of various systems that use the same procedures and conditions.

On the internet, there is a large choice of software system tools that can be obtained free and easily for researching CPU benchmarking. With the tools, the users can run it on their computer and collect the obtained data from public CPU displays. This project will be split into Part 1 and Part 2, where the former will let the students explore CPU benchmark tools, and an assembly program had to be coded for calculating the given mathematical equations by CPU clock support for the latter.

The videos provided in Appendix A are a must to watch for us to perform the CPU benchmarking. The tools that were instructed to be used are CPU-Z, GeekBench, MiniTool Partition Wizard Free 12.8, and the local Task Manager. All of the tools listed are compatible with Windows OS, which will also be the requirement for two of the chosen computers or laptops. The students should download the programs on their own and run the benchmarking test on the selected laptops. The data will be summarised and noted down for public demonstration.

Aim

The main intention of this project is to compare the results of the benchmarking test that has been run on two different computers using 4 different tools. Furthermore, the CPU runtime on the two different devices will be calculated and tested by running an assembly program.

Objectives

- Benchmark both selected laptops with CPU-Z, GeekBench, MiniTool Partition Wizard Free 12.8, and Task Manager.
- Make analysis and comparisons based on the benchmarking results to learn about your CPU's strengths and limitations in different computing activities.
- Run the written assembly program on two of the devices and examine their efficiency based on the CPU runtime.

Scopes (computer specification & benchmarking tools features)

Here are the details of the selected laptops for the benchmarking test.

Specifications	Computer Type	
	Acer Nitro 5	Redmi Book Pro 14 2022
Processor Type	Intel Core i5-12450H	AMD Ryzen 5 6600H
RAM Size	16 GB	16 GB
GPU Type	NVIDIA GeForce RTX 3050 Laptop GPU	AMD Radeon (TM) Graphics
Motherboard	ADL Jimny_ADH	TIMI TM2107
Bus Specs.	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)

For this project, CPU-Z, GeekBench, MiniTool Partition Wizard Free 12.8, and Task Manager will be used to benchmark the laptops.

- CPU-Z is used for obtaining detailed CPU information and performing simple benchmarking on both devices.
 - Processor name and number, codename, process, package, cache levels.
 - Mainboard and chipset.
 - Memory type, size, timings, and module specifications (SPD).
 - Real-time measurement of each core's internal frequency, and memory frequency.
 - CPU-Z is fully supported on Windows® 11.
- GeekBench provides a cross-platform CPU benchmarking.
 - CPU stress test

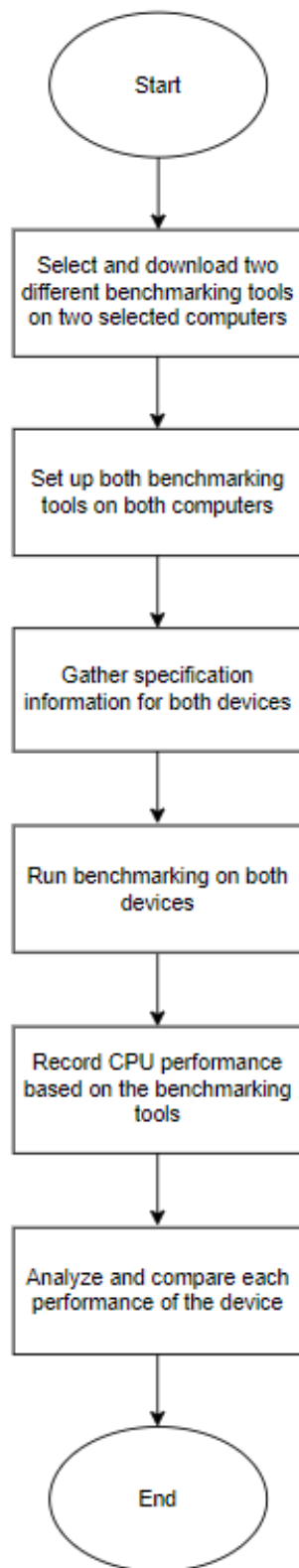
- Single-core and Multi-core performance
- MiniTool Partition Wizard Free 12.8 is used to perform Disk Benchmarking, where it will measure the transfer (read & write) speed under various disk access scenarios like sequential and random.
- Task Manager provides the simulation of the impact of different workloads. The workloads tested are High CPU Load, Mixed CPU Load, and Low CPU Load. The performance of CPU and memory will be taken.

2.0 Part 1

2.1 Overview

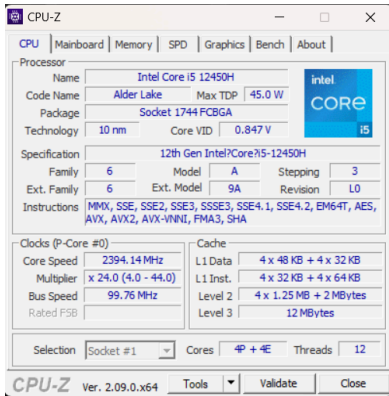
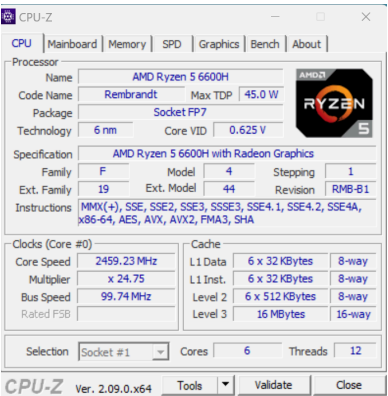
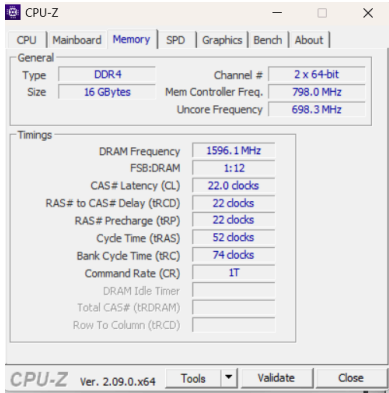
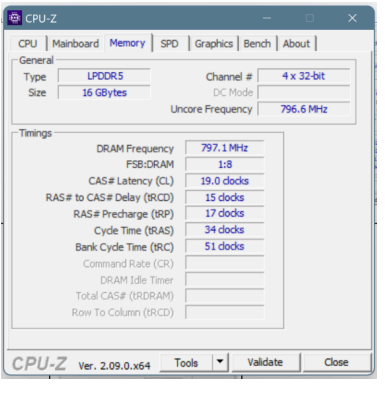
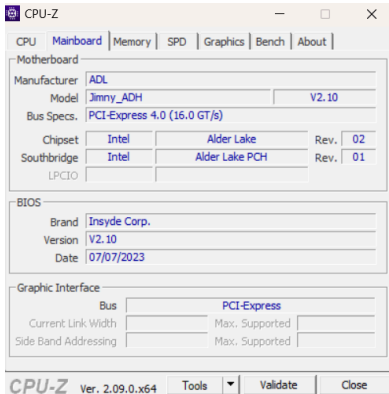
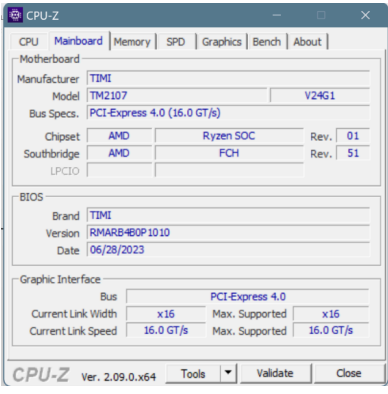
For Part 1, we select four benchmarking tools which are CPU-Z, Geekbench, Task Manager and MiniTool to investigate the performance of two computers and make an analysis and comparison between their performance.

2.2 The Flowchart of the Project



2.3 Benchmarking Results

CPU-Z

	Acer Nitro 5	Redmi Book Pro 14 2022
CPU processor / Cache		
Memory		
Motherboard		

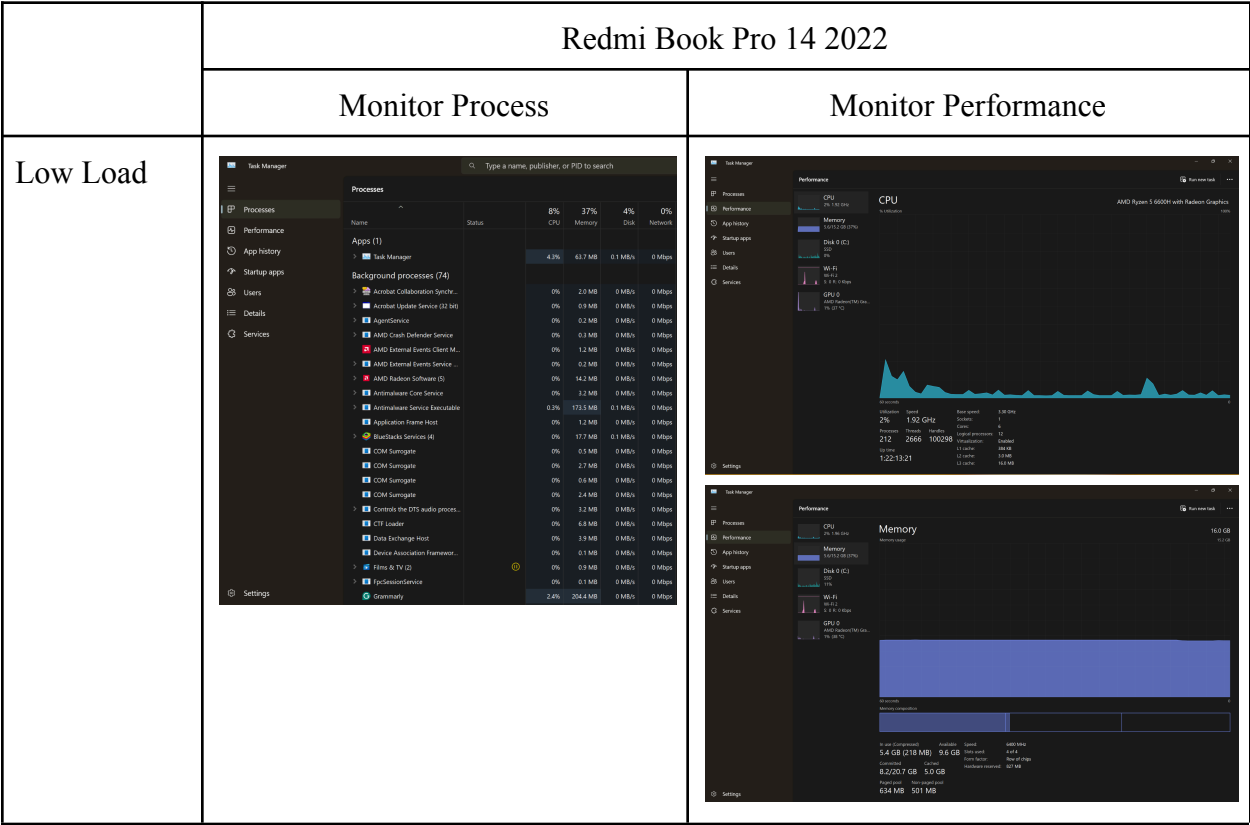
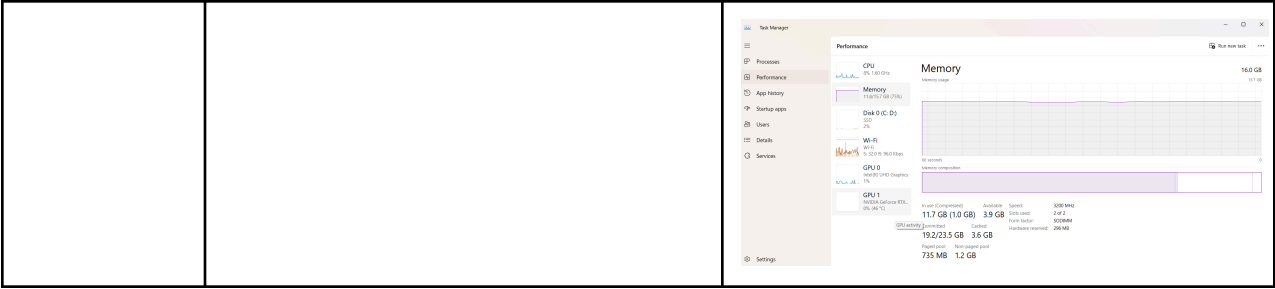
Geekbench

	Acer Nitro 5	Redmi Book Pro 14 2022
CPU Information / Memory Information-	Geekbench 6 CPU GPU Purchase Unlock Help News Benchmark Upgrade Help System Information Operating System: Microsoft Windows 11 Home Single Language (64-bit) Model: Acer Nitro AN515-58 Motherboard: ADL Jimmy_ADH Memory: 16.0 GB DDR4 SDRAM CPU Information Name: Intel Core i5-12450H 1 Processor, 8 Cores, 12 Threads Codename: Alder Lake Package: Socket 1744 FCBGA Base Frequency: 2.50 GHz Maximum Frequency: 4389 MHz CPU Benchmark Geekbench measures CPU performance for everyday tasks using tests that model real-world applications. The CPU Benchmark takes several minutes to complete. Run CPU Benchmark GPU Benchmark GPU API: OpenCL GPU: NVIDIA GeForce RTX 3050 Laptop GPU Geekbench measures GPU compute performance for everyday tasks, including machine learning and computer vision, using tests that model real-world applications. The GPU Benchmark takes several minutes to complete. Run GPU Benchmark	Geekbench 6 CPU GPU Purchase Unlock Help News Benchmark Upgrade Help System Information Operating System: Microsoft Windows 11 Home (64-bit) Model: TIMI Redmi Book Pro 14 2022 Motherboard: TIMI TM2107 Memory: 15.2 GB DDR5 SDRAM CPU Information Name: AMD Ryzen 5 6600H 1 Processor, 6 Cores, 12 Threads Codename: Rembrandt Package: Socket FP7 Base Frequency: 3.30 GHz Maximum Frequency: 2465 MHz CPU Benchmark Geekbench measures CPU performance for everyday tasks using tests that model real-world applications. The CPU Benchmark takes several minutes to complete. Run CPU Benchmark GPU Benchmark GPU API: OpenCL GPU: AMD Radeon(TM) Graphics Geekbench measures GPU compute performance for everyday tasks, including machine learning and computer vision, using tests that model real-world applications. The GPU Benchmark takes several minutes to complete. Run GPU Benchmark
	Acer Nitro AN515-58 1076 Single-Core Score 5547 Multi-Core Score Geekbench 6.3.0 for Windows AVX2 Valid Result Information Upload Date June 29 2024 03:34 AM Views 4 System Information System Information Operating System Microsoft Windows 11 Home Single Language (64-bit) Model Acer Nitro AN515-58 Motherboard ADL Jimmy_ADH Power Plan Acer CPU Information Name Intel Core i5-12450H Topology 1 Processor, 8 Cores, 12 Threads Identifier GenuineIntel Family 6 Model 154 Stepping 3 Base Frequency 2.50 GHz Cluster 1 4 Cores Cluster 2 4 Cores Maximum Frequency 4389 MHz Package Socket 1744 FCBGA Codename Alder Lake L1 Instruction Cache 32.0 KB x 6 L1 Data Cache 48.0 KB x 6 L2 Cache 1.25 MB x 1 L3 Cache 12.0 MB x 1 Memory Information Size 16.00 GB Transfer Rate 3192 MT/s Type DDR4 SDRAM Channels 2	TIMI Redmi Book Pro 14 2022 1069 Single-Core Score 4555 Multi-Core Score Geekbench 6.3.0 for Windows AVX2 Valid Result Information Upload Date June 29 2024 03:09 PM Views 1 System Information System Information Operating System Microsoft Windows 11 Home (64-bit) Model TIMI Redmi Book Pro 14 2022 Motherboard TIMI TM2107 Power Plan Balanced CPU Information Name AMD Ryzen 5 6600H Topology 1 Processor, 6 Cores, 12 Threads Identifier AuthenticAMD Family 25 Model 68 Stepping 1 Base Frequency 3.30 GHz Cluster 1 6 Cores Maximum Frequency 2465 MHz Package Socket FP7 Codename Rembrandt L1 Instruction Cache 32.0 KB x 6 L1 Data Cache 32.0 KB x 6 L2 Cache 512 KB x 6 L3 Cache 16.0 MB x 1 Memory Information Size 15.19 GB Transfer Rate 1592 MT/s Type DDR5 SDRAM Channels 4

Single-Core Performance	<table><tr><th colspan="2">Single-Core Performance</th></tr><tr><th>Single-Core Score</th><th>1076</th></tr><tr><td>File Compression</td><td>1111 159.6 MB/sec</td></tr><tr><td>Navigation</td><td>913 5.50 routes/sec</td></tr><tr><td>HTML5 Browser</td><td>1201 24.6 pages/sec</td></tr><tr><td>PDF Renderer</td><td>1022 23.6 Mpixels/sec</td></tr><tr><td>Photo Library</td><td>1213 16.5 images/sec</td></tr><tr><td>Clang</td><td>1215 5.98 Klines/sec</td></tr><tr><td>Text Processing</td><td>1210 96.9 pages/sec</td></tr><tr><td>Asset Compression</td><td>1041 32.2 MB/sec</td></tr><tr><td>Object Detection</td><td>1021 30.5 images/sec</td></tr><tr><td>Background Blur</td><td>1345 5.57 images/sec</td></tr><tr><td>Horizon Detection</td><td>1111 34.6 Mpixels/sec</td></tr><tr><td>Object Remover</td><td>810 62.3 Mpixels/sec</td></tr><tr><td>HDR</td><td>911 26.7 Mpixels/sec</td></tr><tr><td>-</td><td>1203</td></tr><tr><td>Photo Filter</td><td>11.9 images/sec</td></tr><tr><td>Ray Tracer</td><td>951 920.1 Kpixels/sec</td></tr><tr><td>Structure from Motion</td><td>996 31.5 Kpixels/sec</td></tr></table>	Single-Core Performance		Single-Core Score	1076	File Compression	1111 159.6 MB/sec	Navigation	913 5.50 routes/sec	HTML5 Browser	1201 24.6 pages/sec	PDF Renderer	1022 23.6 Mpixels/sec	Photo Library	1213 16.5 images/sec	Clang	1215 5.98 Klines/sec	Text Processing	1210 96.9 pages/sec	Asset Compression	1041 32.2 MB/sec	Object Detection	1021 30.5 images/sec	Background Blur	1345 5.57 images/sec	Horizon Detection	1111 34.6 Mpixels/sec	Object Remover	810 62.3 Mpixels/sec	HDR	911 26.7 Mpixels/sec	-	1203	Photo Filter	11.9 images/sec	Ray Tracer	951 920.1 Kpixels/sec	Structure from Motion	996 31.5 Kpixels/sec	<table><tr><th colspan="2">Single-Core Performance</th></tr><tr><th>Single-Core Score</th><th>1069</th></tr><tr><td>File Compression</td><td>1196 171.7 MB/sec</td></tr><tr><td>Navigation</td><td>1085 6.54 routes/sec</td></tr><tr><td>HTML5 Browser</td><td>972 19.9 pages/sec</td></tr><tr><td>PDF Renderer</td><td>1083 25.0 Mpixels/sec</td></tr><tr><td>Photo Library</td><td>997 13.5 images/sec</td></tr><tr><td>Clang</td><td>1029 5.07 Klines/sec</td></tr><tr><td>Text Processing</td><td>967 77.4 pages/sec</td></tr><tr><td>Asset Compression</td><td>1113 34.5 MB/sec</td></tr><tr><td>Object Detection</td><td>564 16.9 images/sec</td></tr><tr><td>Background Blur</td><td>1548 6.41 images/sec</td></tr><tr><td>Horizon Detection</td><td>1496 46.6 Mpixels/sec</td></tr><tr><td>Object Remover</td><td>1111 85.5 Mpixels/sec</td></tr><tr><td>HDR</td><td>1290 37.8 Mpixels/sec</td></tr><tr><td>Photo Filter</td><td>1395 13.8 images/sec</td></tr><tr><td>Ray Tracer</td><td>1006 973.1 Kpixels/sec</td></tr><tr><td>Structure from Motion</td><td>1031 32.6 Kpixels/sec</td></tr></table>	Single-Core Performance		Single-Core Score	1069	File Compression	1196 171.7 MB/sec	Navigation	1085 6.54 routes/sec	HTML5 Browser	972 19.9 pages/sec	PDF Renderer	1083 25.0 Mpixels/sec	Photo Library	997 13.5 images/sec	Clang	1029 5.07 Klines/sec	Text Processing	967 77.4 pages/sec	Asset Compression	1113 34.5 MB/sec	Object Detection	564 16.9 images/sec	Background Blur	1548 6.41 images/sec	Horizon Detection	1496 46.6 Mpixels/sec	Object Remover	1111 85.5 Mpixels/sec	HDR	1290 37.8 Mpixels/sec	Photo Filter	1395 13.8 images/sec	Ray Tracer	1006 973.1 Kpixels/sec	Structure from Motion	1031 32.6 Kpixels/sec
Single-Core Performance																																																																												
Single-Core Score	1076																																																																											
File Compression	1111 159.6 MB/sec																																																																											
Navigation	913 5.50 routes/sec																																																																											
HTML5 Browser	1201 24.6 pages/sec																																																																											
PDF Renderer	1022 23.6 Mpixels/sec																																																																											
Photo Library	1213 16.5 images/sec																																																																											
Clang	1215 5.98 Klines/sec																																																																											
Text Processing	1210 96.9 pages/sec																																																																											
Asset Compression	1041 32.2 MB/sec																																																																											
Object Detection	1021 30.5 images/sec																																																																											
Background Blur	1345 5.57 images/sec																																																																											
Horizon Detection	1111 34.6 Mpixels/sec																																																																											
Object Remover	810 62.3 Mpixels/sec																																																																											
HDR	911 26.7 Mpixels/sec																																																																											
-	1203																																																																											
Photo Filter	11.9 images/sec																																																																											
Ray Tracer	951 920.1 Kpixels/sec																																																																											
Structure from Motion	996 31.5 Kpixels/sec																																																																											
Single-Core Performance																																																																												
Single-Core Score	1069																																																																											
File Compression	1196 171.7 MB/sec																																																																											
Navigation	1085 6.54 routes/sec																																																																											
HTML5 Browser	972 19.9 pages/sec																																																																											
PDF Renderer	1083 25.0 Mpixels/sec																																																																											
Photo Library	997 13.5 images/sec																																																																											
Clang	1029 5.07 Klines/sec																																																																											
Text Processing	967 77.4 pages/sec																																																																											
Asset Compression	1113 34.5 MB/sec																																																																											
Object Detection	564 16.9 images/sec																																																																											
Background Blur	1548 6.41 images/sec																																																																											
Horizon Detection	1496 46.6 Mpixels/sec																																																																											
Object Remover	1111 85.5 Mpixels/sec																																																																											
HDR	1290 37.8 Mpixels/sec																																																																											
Photo Filter	1395 13.8 images/sec																																																																											
Ray Tracer	1006 973.1 Kpixels/sec																																																																											
Structure from Motion	1031 32.6 Kpixels/sec																																																																											
Multiple-Core Performance	<table><tr><th colspan="2">Multi-Core Performance</th></tr><tr><th>Multi-Core Score</th><th>5547</th></tr><tr><td>File Compression</td><td>3576 513.5 MB/sec</td></tr><tr><td>Navigation</td><td>7046 42.5 routes/sec</td></tr><tr><td>HTML5 Browser</td><td>6165 126.2 pages/sec</td></tr><tr><td>PDF Renderer</td><td>6113 141.0 Mpixels/sec</td></tr><tr><td>Photo Library</td><td>6724 91.3 images/sec</td></tr><tr><td>Clang</td><td>7513 37.0 Klines/sec</td></tr><tr><td>Text Processing</td><td>1338 107.2 pages/sec</td></tr><tr><td>Asset Compression</td><td>8562 265.3 MB/sec</td></tr><tr><td>Object Detection</td><td>3903 116.8 images/sec</td></tr><tr><td>Background Blur</td><td>6793 28.1 images/sec</td></tr><tr><td>Horizon Detection</td><td>8664 269.6 Mpixels/sec</td></tr><tr><td>Object Remover</td><td>5328 409.7 Mpixels/sec</td></tr><tr><td>HDR</td><td>6030 177.0 Mpixels/sec</td></tr><tr><td>Photo Filter</td><td>5264 52.2 images/sec</td></tr><tr><td>Ray Tracer</td><td>7481 7.24 Mpixels/sec</td></tr><tr><td>Structure from Motion</td><td>7316 231.6 Kpixels/sec</td></tr></table>	Multi-Core Performance		Multi-Core Score	5547	File Compression	3576 513.5 MB/sec	Navigation	7046 42.5 routes/sec	HTML5 Browser	6165 126.2 pages/sec	PDF Renderer	6113 141.0 Mpixels/sec	Photo Library	6724 91.3 images/sec	Clang	7513 37.0 Klines/sec	Text Processing	1338 107.2 pages/sec	Asset Compression	8562 265.3 MB/sec	Object Detection	3903 116.8 images/sec	Background Blur	6793 28.1 images/sec	Horizon Detection	8664 269.6 Mpixels/sec	Object Remover	5328 409.7 Mpixels/sec	HDR	6030 177.0 Mpixels/sec	Photo Filter	5264 52.2 images/sec	Ray Tracer	7481 7.24 Mpixels/sec	Structure from Motion	7316 231.6 Kpixels/sec	<table><tr><th colspan="2">Multi-Core Performance</th></tr><tr><th>Multi-Core Score</th><th>4555</th></tr><tr><td>File Compression</td><td>4282 614.9 MB/sec</td></tr><tr><td>Navigation</td><td>5333 32.1 routes/sec</td></tr><tr><td>HTML5 Browser</td><td>3918 80.2 pages/sec</td></tr><tr><td>PDF Renderer</td><td>5272 121.6 Mpixels/sec</td></tr><tr><td>Photo Library</td><td>5521 74.9 images/sec</td></tr><tr><td>Clang</td><td>6993 34.4 Klines/sec</td></tr><tr><td>Text Processing</td><td>1296 103.8 pages/sec</td></tr><tr><td>Asset Compression</td><td>7670 237.7 MB/sec</td></tr><tr><td>Object Detection</td><td>2297 68.7 images/sec</td></tr><tr><td>Background Blur</td><td>5916 24.5 images/sec</td></tr><tr><td>Horizon Detection</td><td>5237 163.0 Mpixels/sec</td></tr><tr><td>Object Remover</td><td>5269 405.1 Mpixels/sec</td></tr><tr><td>HDR</td><td>4602 135.1 Mpixels/sec</td></tr><tr><td>Photo Filter</td><td>3227 32.0 images/sec</td></tr><tr><td>Ray Tracer</td><td>9042 8.75 Mpixels/sec</td></tr><tr><td>Structure from Motion</td><td>5172 163.8 Kpixels/sec</td></tr></table>	Multi-Core Performance		Multi-Core Score	4555	File Compression	4282 614.9 MB/sec	Navigation	5333 32.1 routes/sec	HTML5 Browser	3918 80.2 pages/sec	PDF Renderer	5272 121.6 Mpixels/sec	Photo Library	5521 74.9 images/sec	Clang	6993 34.4 Klines/sec	Text Processing	1296 103.8 pages/sec	Asset Compression	7670 237.7 MB/sec	Object Detection	2297 68.7 images/sec	Background Blur	5916 24.5 images/sec	Horizon Detection	5237 163.0 Mpixels/sec	Object Remover	5269 405.1 Mpixels/sec	HDR	4602 135.1 Mpixels/sec	Photo Filter	3227 32.0 images/sec	Ray Tracer	9042 8.75 Mpixels/sec	Structure from Motion	5172 163.8 Kpixels/sec		
Multi-Core Performance																																																																												
Multi-Core Score	5547																																																																											
File Compression	3576 513.5 MB/sec																																																																											
Navigation	7046 42.5 routes/sec																																																																											
HTML5 Browser	6165 126.2 pages/sec																																																																											
PDF Renderer	6113 141.0 Mpixels/sec																																																																											
Photo Library	6724 91.3 images/sec																																																																											
Clang	7513 37.0 Klines/sec																																																																											
Text Processing	1338 107.2 pages/sec																																																																											
Asset Compression	8562 265.3 MB/sec																																																																											
Object Detection	3903 116.8 images/sec																																																																											
Background Blur	6793 28.1 images/sec																																																																											
Horizon Detection	8664 269.6 Mpixels/sec																																																																											
Object Remover	5328 409.7 Mpixels/sec																																																																											
HDR	6030 177.0 Mpixels/sec																																																																											
Photo Filter	5264 52.2 images/sec																																																																											
Ray Tracer	7481 7.24 Mpixels/sec																																																																											
Structure from Motion	7316 231.6 Kpixels/sec																																																																											
Multi-Core Performance																																																																												
Multi-Core Score	4555																																																																											
File Compression	4282 614.9 MB/sec																																																																											
Navigation	5333 32.1 routes/sec																																																																											
HTML5 Browser	3918 80.2 pages/sec																																																																											
PDF Renderer	5272 121.6 Mpixels/sec																																																																											
Photo Library	5521 74.9 images/sec																																																																											
Clang	6993 34.4 Klines/sec																																																																											
Text Processing	1296 103.8 pages/sec																																																																											
Asset Compression	7670 237.7 MB/sec																																																																											
Object Detection	2297 68.7 images/sec																																																																											
Background Blur	5916 24.5 images/sec																																																																											
Horizon Detection	5237 163.0 Mpixels/sec																																																																											
Object Remover	5269 405.1 Mpixels/sec																																																																											
HDR	4602 135.1 Mpixels/sec																																																																											
Photo Filter	3227 32.0 images/sec																																																																											
Ray Tracer	9042 8.75 Mpixels/sec																																																																											
Structure from Motion	5172 163.8 Kpixels/sec																																																																											

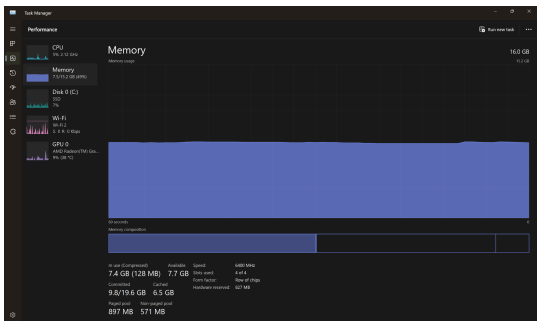
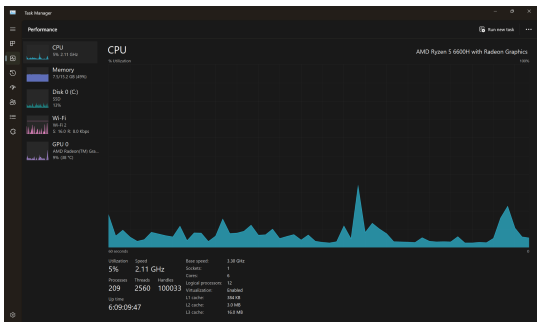
Monitor Task Manager

[illegible]



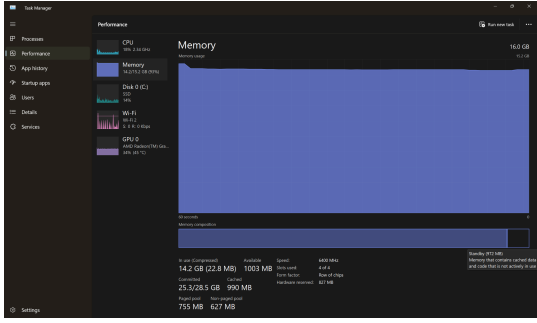
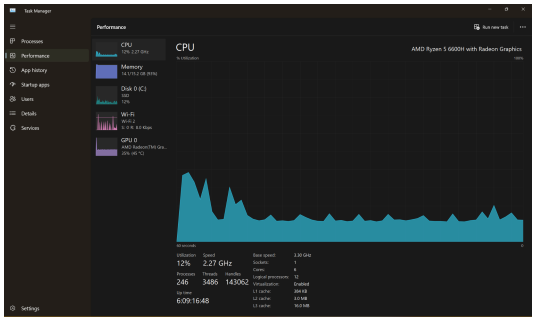
Middle Load

Processes					6%	49%	12%	0%
					CPU	Memory	Disk	Network
Apps (2)								
Google Chrome (16)					2.4%	1,535.0 MB	0.2 MB/s	0 Mbps
Task Manager					1.0%	57.0 MB	0.1 MB/s	0 Mbps
Background processes (65)								
Account Collaboration Synch...					0%	6.9 MB	0 MB/s	0 Mbps
Account Update Service (32 bit)					0%	0.1 MB	0 MB/s	0 Mbps
AgentService					0%	3.6 MB	0 MB/s	0 Mbps
AMD Crash Defender Service					0%	0.4 MB	0 MB/s	0 Mbps
AMD External Events Client M...					0%	1.6 MB	0 MB/s	0 Mbps
AMD External Events Service ...					0%	0.3 MB	0 MB/s	0 Mbps
AMD Radeon Software (5)					0%	12.4 MB	0 MB/s	0 Mbps
Antimalware Core Service					0%	7.7 MB	0 MB/s	0 Mbps
Antimalware Service Executable					0.3%	164.5 MB	0.1 MB/s	0 Mbps
Application Frame Host					0%	5.2 MB	0 MB/s	0 Mbps
BlueStacks Services (4)					0%	36.6 MB	0 MB/s	0 Mbps
COM Surrogate					0%	0.5 MB	0 MB/s	0 Mbps
COM Surrogate					0%	1.8 MB	0 MB/s	0 Mbps
Controls the DTS audio proces...					0%	1.2 MB	0 MB/s	0 Mbps
CTF Loader					0%	3.4 MB	0 MB/s	0 Mbps
FpcSessionService					0%	0.1 MB	0 MB/s	0 Mbps
Grammarly					0.2%	85.3 MB	0 MB/s	0 Mbps
Host Process for Windows Tas...					0%	2.3 MB	0 MB/s	0 Mbps
Host Process for Windows Tas...					0%	3.1 MB	0 MB/s	0 Mbps
IpphaseScanOccorService					0%	2.1 MB	0 MB/s	0 Mbps
Hyper-V Host Compute Service					0%	1.2 MB	0 MB/s	0 Mbps

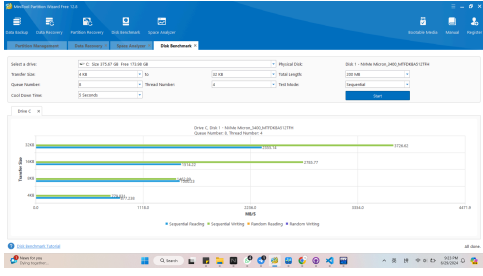
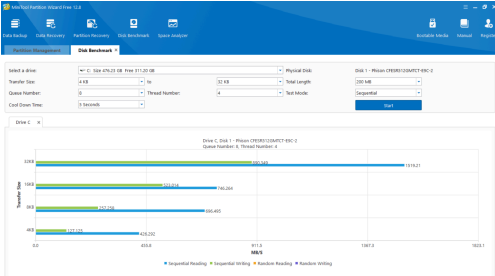
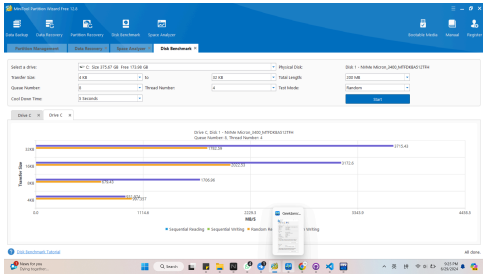
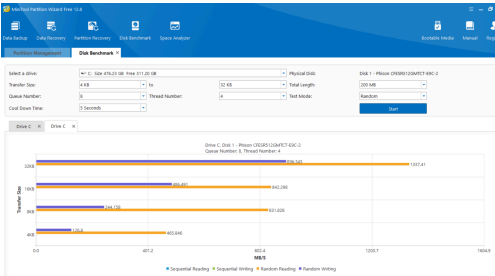


High Load

Processes					14%	94%	9%	0%
					CPU	Memory	Disk	Network
Apps (12)								
BlueStacks					7.8%	940.9 MB	0 MB/s	0 Mbps
DellEMC_0.9.13_x64					0%	132.8 MB	0 MB/s	0 Mbps
Google Chrome (16)					0.3%	1,108.8 MB	0 MB/s	0 Mbps
Intel® Digital Publishing Application					0%	482.5 MB	0 MB/s	0 Mbps
Microsoft Store (2)					0%	2.6 MB	0 MB/s	0 Mbps
mspaint.exe					0%	49.6 MB	0 MB/s	0 Mbps
Photos					0%	129.9 MB	0 MB/s	0 Mbps
Task Manager					1.4%	60.0 MB	0.1 MB/s	0 Mbps
Visual Studio Code (9)					0%	588.8 MB	0 MB/s	0 Mbps
VisualStudio-M					0.2%	13.9 MB	0 MB/s	0 Mbps
VisualStudio					0%	1,742.7 MB	0 MB/s	0 Mbps
Windows Explorer					0%	181.5 MB	0 MB/s	0 Mbps
Background processes (74)								
Account Collaboration Synch...					0%	2.1 MB	0 MB/s	0 Mbps
Account Update Service (32 bit)					0%	0.1 MB	0 MB/s	0 Mbps
AgentService					0%	0.3 MB	0 MB/s	0 Mbps
AMD Crash Defender Service					0%	0.1 MB	0 MB/s	0 Mbps
AMD External Events Client M...					0%	1.0 MB	0 MB/s	0 Mbps
AMD External Events Service ...					0%	0.1 MB	0 MB/s	0 Mbps
AMD Radeon Software (5)					0%	8.0 MB	0 MB/s	0 Mbps
Antimalware Core Service					0%	5.0 MB	0 MB/s	0 Mbps
Antimalware Service Executable					0%	161.9 MB	0 MB/s	0 Mbps
Application Frame Host					0%	10.4 MB	0 MB/s	0 Mbps
BlueStacks Server Interface					0%	2.6 MB	0 MB/s	0 Mbps



MiniTool Partition Wizard

	Acer Nitro 5	Redmi Book Pro 14 2022
Sequential		
Random		

2.4 Analysis, Comparison, Discussion

After running the benchmark test on two devices, we get the required performance data. By using the collected data, we can obtain information about the CPU performance.

CPU-Z

Benchmarking	Computer Type	
	Acer Nitro 5	Redmi Book Pro 14 2022
CPU Technology	10 nm	6 nm
Number of Core	8	6
Core Speed	2394.14 MHz	2459.23 MHz
Bus Speed	99.76 MHz	99.74 MHz
Cache L1 Data	4 x 48 KB + 4 x 32 KB	6 x 32 KB 8-way
Cache L1 Insta.	4 x 32 KB + 4 x 64 KB	6 x 32 KB 8-way
Cache Level 2	4 x 1.25 MB + 2 MB	6 x 512 KB 8-way
Cache Level 3	12 MB	16 MB 16-way
Memory Type	DDR 4	LPDDR5
RAM Size	16 GB	16 GB
DRAM Frequency	1596.1 MHz	797.1 MHz
Bus Specs.	PCI-Express 4.0 (16.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)
Chipset	Intel Alder Lake	AMD Ryzen SOC
Bios Version	V2. 10	RMARB4B0P 1010

CPU Technology and Number of Cores:

Redmi Book Pro 14 2022 uses a 6 nm technology with 6 cores, suggesting more sophisticated manufacturing technology but fewer cores, while Acer Nitro 5 uses a 10 nm technology with 8 cores, suggesting potentially superior power efficiency and multitasking capacity.

Core Speed and Bus Speed:

Redmi Book Pro 14 2022 has a slightly higher core speed of 2459.23 MHz but a similar bus speed of 99.74 MHz, while Acer Nitro 5 has a core speed of 2394.14 MHz and a bus speed of 99.76 MHz.

Cache Memory:

In cache L1 data and instruction, the Redmi Book Pro 14 2022 has consistent 32 KB 8-way L1 caches, however the Acer Nitro 5 has a more intricate cache configuration with variable sizes.

In cache levels 2 and 3, the Redmi Book Pro 14 2022 may run better in apps that rely on cache because it has a larger L3 cache (16 MB) than the Acer Nitro 5 (12 MB).

Memory Type, RAM Size and DRAM Frequency:

Both laptops have 16 GB of RAM, however the Redmi Book Pro 14 2022 uses LPDDR5 at 797.1 MHz, while the Acer Nitro 5 uses DDR4 at a higher frequency of 1596.1 MHz. An Acer Nitro 5 with a higher DRAM frequency might perform better when doing memory-intensive tasks.

Chipset and BIOS:

The AMD Ryzen SOC, which is likewise renowned for its strong performance, powers the Redmi Book Pro 14 2022, while the Intel Alder Lake chipset, which powers the Acer Nitro 5, is well known for its efficiency and performance cores. The firmware support level and potential optimization for each device are indicated by the distinct BIOS versions.

Conclusion for comparing both devices using CPU-Z:

In conclusion, due to its higher core count, faster DRAM frequency, and larger L1 caches, the Acer Nitro 5 seems to perform better in multitasking and memory-intensive applications, while the Redmi Book Pro 14 2022 benefits from a more advanced manufacturing process (6 nm), a larger L3 cache, and possibly better power efficiency because of its LPDDR5 memory.

Geekbench

Single-Core Performance

Task	Acer Nitro 5	Redmi Book Pro 14 2022
File Compression	1111159.6 MB/sec	1196171.7 MB/sec
Navigation	9135.50 routes/sec	10856.54 routes/sec
HTML5 Browser	120124.6 pages/sec	97219.9 pages/sec
Text Processing	121096.9 pages/sec	96777.4 pages/sec
Object Detection	102130.5 images/sec	56416.9 images/sec
Structure from Motion	99631.5 Kpixels/sec	103132.6 Kpixels/sec

Text processing, object detection, and HTML5 browsing are three areas where the Acer Nitro 5 excels. These skills are essential for a variety of applications, from everyday productivity to more specialist jobs like machine learning. Improved file compression, navigation, and structure from motion performance show that the Redmi Book Pro 14 2022 is more effective at certain activities requiring quick data processing and intricate calculations.

Multiple-Core Performance

Task	Acer Nitro 5	Redmi Book Pro 14 2022
File Compression	3576513.5 MB/sec	4282614.9 MB/sec
Navigation	704642.5 routes/sec	533332.1 routes/sec
HTML5 Browser	6165126.2 pages/sec	39180.2 pages/sec
Text Processing	1338107.2 pages/sec	1296103.8 pages/sec
Object Detection	3903116.8 images/sec	229768.7 images/sec
Structure from Motion	7316231.6 Kpixels/sec	5172163.8 Kpixels/sec

In terms of navigation, HTML5 browsing, text processing, object detection, and structure from motion, the Acer Nitro 5 performs better. This suggests good performance in a variety of applications, including computer vision and 3D reconstruction, as well as general productivity.

Better file compression results indicate that the Redmi Book Pro 14 2022 is effective at activities requiring quick data compression and decompression.

Monitor Task Manager

Low Load (memory consumption around 30% to 50%)

		Acer Nitro 5	Redmi Book Pro 14 2022
Monitor Process	CPU Usage	3%	8%
	Disk Usage	1%	4%
Monitor Performance	CPU Speed	1.32 GHz	1.92 GHz
	CPU Processes	242	212
	CPU Threads	2883	2666
	Memory in use (Compressed)	6.3 GB (480 MB)	5.4 GB (218 MB)
	Memory Available	9.2 GB	9.6 GB
	Cached	5.1 GB	5.0 GB

Monitor Process:

Under low load conditions, the Redmi Book Pro 14 2022 exhibits a greater CPU utilization (8%) than the Acer Nitro 5. This suggests that the Acer Nitro 5 may manage CPU resources for light tasks more effectively. The Acer Nitro 5 uses 1% of its disk space, whereas the Redmi Book Pro 14 2022 uses 4%. This implies that when under low load, the Acer Nitro 5 manages disk activity more effectively.

Monitor Performance:

The difference between the CPU speeds of the Redmi Book Pro 14 2022 and the Acer Nitro 5 is 1.92 GHz compared to 1.32 GHz, respectively. For this reason, simple activities may be responded by the Redmi Book Pro quickly. Acer Nitro 5 has 242 processes and 2883 threads in relation to CPU processes and threads, while Redmi Book Pro 14 of 2022 has 212 processes and 2666 threads. Even when the CPU usage is low, this suggests that the Acer Nitro 5 can handle

more tasks simultaneously. The Acer Nitro 5 uses 6.3 GB of RAM in memory in use, while the Redmi Book Pro 14 2022 utilizes 5.4 GB. Furthermore, the compressed memory of the Acer Nitro 5 (480 MB) is greater than that of the Redmi Book Pro 14 2022 (218 MB), indicating that the Acer system might be processing more data that has been cached. With 9.6 GB of accessible memory—a little bit more than the 9.2 GB of memory on the Acer Nitro 5—the Redmi Book Pro 14 2022 has a little bit extra RAM that can be used for various tasks. Cache memory for both laptops is nearly equal—5.1 GB on the Acer Nitro 5 and 5.0 GB on the Redmi Book Pro 14 2022.

Middle Load (memory consumption around 50 % to 60%)

		Acer Nitro 5	Redmi Book Pro 14 2022
Monitor Process	CPU Usage	2%	6%
	Disk Usage	1%	12%
Monitor Performance	CPU Speed	1.37 GHz	2.11 GHz
	CPU Processes	267	209
	CPU Threads	3461	2560
	Memory in use (Compressed)	8.4 GB (702 MB)	7.4 GB (128 MB)
	Memory Available	7.1 GB	7.7 GB
	Cached	5.4 GB	6.5 GB

Monitor Process:

In conditions of moderate load, the Redmi Book Pro 14 2022 has a greater CPU utilization (6%) in comparison to the Acer Nitro 5. This implies that the Acer Nitro 5 manages CPU resources more effectively. Disk consumption on the Acer Nitro 5 is notably lower (1%) than on the Redmi Book Pro 14 2022 (12%). This suggests that the Acer Nitro 5 has improved disk activity control at intermediate load.

Monitor Performance:

In terms of CPU speed, the Redmi Book Pro 14 2022 outperforms the Acer Nitro 5 (1.37 GHz) with a faster CPU (2.11 GHz). As a result, the Redmi Book Pro may respond to moderate tasks more quickly. The Redmi Book Pro 14 2022 runs 209 processes and 2560 threads, whereas the Acer Nitro 5 runs 267 processes and 3461 threads in terms of CPU processes and threads. The Redmi Book Pro 14 2022 only uses 7.4 GB of memory, while the Acer Nitro 5 uses 8.4 GB. In addition, the Acer Nitro 5's compressed memory (702 MB) is larger than the Redmi Book Pro 14 2022's (128 MB), suggesting that the Acer system may be processing more cached data. The Redmi Book Pro 14 2022 has 7.7 GB of accessible memory, which is little higher than the Acer Nitro 5's 7.1 GB, indicating a little bit more RAM that can be used for other tasks. With 6.5 GB greater cached memory than the Acer Nitro 5 (5.4 GB), the Redmi Book Pro 14 2022 may operate better when accessing frequently visited material.

High Load (memory consumption above 70% above)

		Acer Nitro 5	Redmi Book Pro 14 2022
Monitor Process	CPU Usage	5%	14%
	Disk Usage	1%	9%
Monitor Performance	CPU Speed	1.18 GHz	2.27 GHz
	CPU Processes	323	246
	CPU Threads	4489	3486
	Memory in use (Compressed)	11.7 GB (1.0 GB)	14.2 GB (22.8 MB)
	Memory Available	3.9 GB	1003 MB
	Cached	3.6 GB	990 MB

Monitor Process:

When under high load, the Redmi Book Pro 14 2022 displays a much greater CPU use (14%) than the Acer Nitro 5 (5%) does. This suggests that, even with high workloads, the Acer Nitro 5 manages CPU resources more effectively.

Disk consumption is lower on the Acer Nitro 5 (1%) than it is on the Redmi Book Pro 14 2022 (9%). This implies that under heavy load, the Acer Nitro 5 manages disk activity more effectively.

Monitor Performance:

Under heavy strain, the Redmi Book Pro 14 2022's CPU runs at 2.27 GHz, while the Acer Nitro 5's runs at 1.18 GHz. This might enable the Redmi Book Pro to do hard tasks more quickly. Compared to the Redmi Book Pro 14 2022, which operates 246 processes and 3486 threads, the Acer Nitro 5 runs 323 processes and 4489 threads. This suggests that under heavy pressure, the Acer Nitro 5 might manage more tasks running simultaneously. In contrast to the Acer Nitro 5, which uses 11.7 GB of RAM, the Redmi Book Pro 14 2022 requires 14.2 GB. However, the Redmi Book Pro 14 2022's compressed memory is much smaller (22.8 MB) than the Acer Nitro 5's (1.0 GB), indicating that the Redmi system compresses memory more effectively. When compared to the Redmi Book Pro 14 2022 (1003 MB), the Acer Nitro 5 has greater accessible memory (3.9 GB), meaning that it can handle more tasks when the load is high. The Redmi Book Pro 14 2022 has 990 MB of cached memory, but the Acer Nitro 5 has 3.6 GB, which could help with often requested data performance.

Conclusion for comparing both laptops using monitor task manager:

All things considered, the Redmi Book Pro 14 2022 displays greater CPU speeds, indicating its performance focus, while the Acer Nitro 5 regularly runs at lower CPU speeds under load, demonstrating effective battery saving. In addition, the Acer Nitro 5 has more cached and available memory under all load scenarios, demonstrating more effective memory management. As the Redmi Book Pro 14 2022 displays higher resource usage and emphasizes performance but with less efficiency in resource management, the Acer Nitro 5 excels in resource efficiency and consistent performance under various loads overall.

MiniTool Partition Wizard

Sequential Mode Test

Transfer Size	Acer Nitro 5		Redmi Book Pro 14 2022	
	Sequential Reading (MB/s)	Sequential Writing (MB/s)	Sequential Reading (MB/s)	Sequential Writing (MB/s)
32 KB	2355.14	3726.62	1519.21	890.349
16 KB	1514.22	2785.77	746.264	523.014
8 KB	1500.23	1462.89	696.495	257.258
4 KB	877.238	779.831	426.292	127.125

In both sequential read and write operations, the Acer Nitro 5 performs noticeably better than the Redmi Book Pro 14 2022 across all transfer sizes. Sequential reading indicates that the Acer Nitro 5 outperforms the Redmi Book Pro 14 2022 in sequential read rates, with the largest disparity occurring at the 32 KB transfer size. Nevertheless, the Acer Nitro 5 demonstrates noticeably faster sequential write speeds as well, particularly at bigger transfer sizes (32 KB and 16 KB). These findings suggest operations requiring substantial data transfers, like file transferring and video editing, would be better performed by the Acer Nitro 5. For high-speed computing requirements, the Acer Nitro 5 is a better option because to its improved performance in sequential reading and writing.

Random Mode Test

Transfer Size	Acer Nitro 5		Redmi Book Pro 14 2022	
	Random Reading (MB/s)	Random Writing (MB/s)	Random Reading (MB/s)	Random Writing (MB/s)
32 KB	1782.59	3715.43	1337.41	896.343
16 KB	2022.53	3172.6	842.290	486.491
8 KB	679.43	1706.96	831.828	244.158
4 KB	997.357	931.974	465.846	126.8

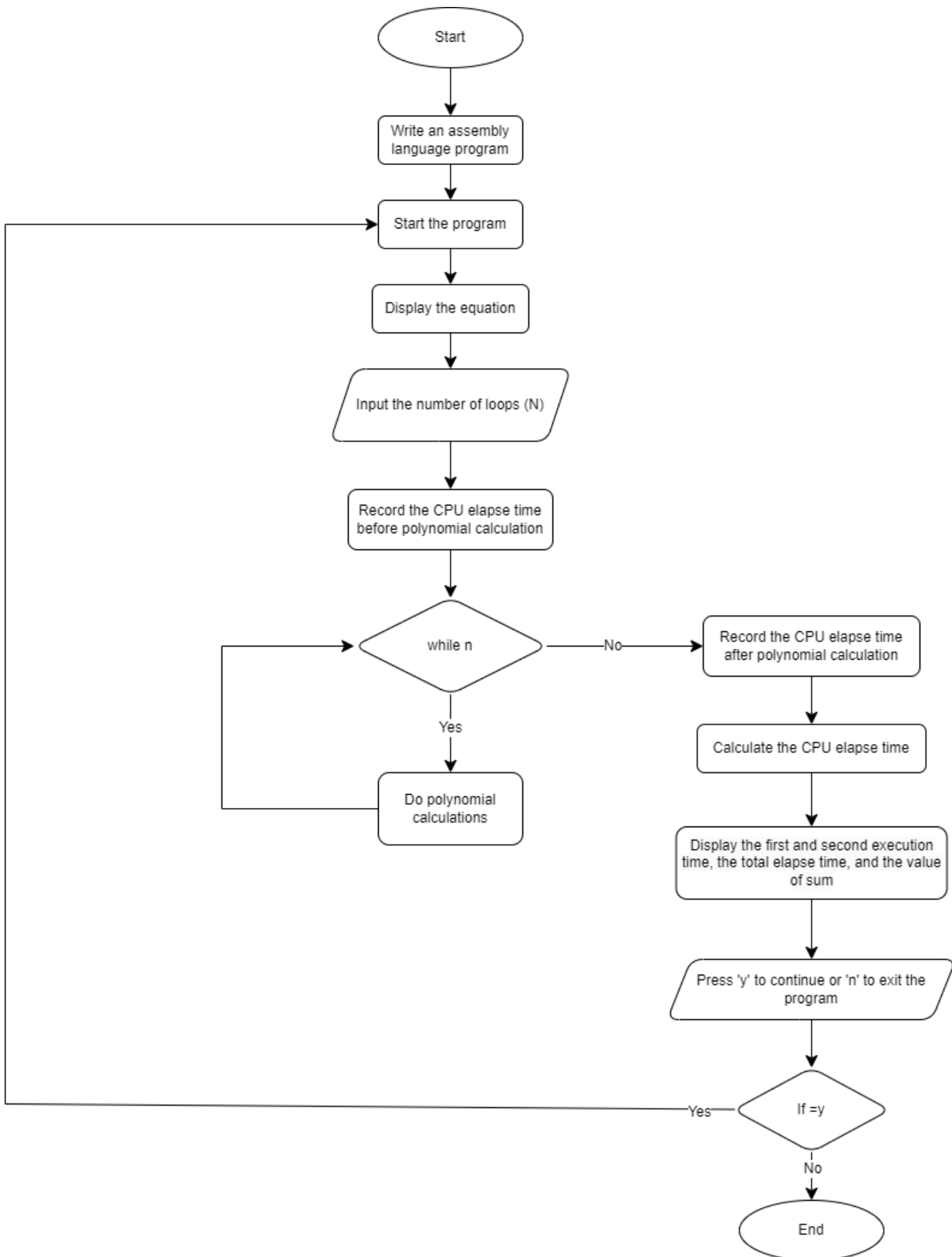
The Acer Nitro 5 exhibits faster random read speeds in random reading with transfer sizes of 32 KB, 16 KB, and 4 KB. At the 8 KB transfer size, the Redmi Book Pro 14 2022 displays a slightly faster random read speed. Comparing the Acer Nitro 5 to the Redmi Book Pro 14 2022, the former shows noticeably faster random write rates across all transfer sizes. In conclusion, for most transfer sizes, the Acer Nitro 5 performs consistently better in random read and write operations than the Redmi Book Pro 14 2022.

3.0 Part 2

3.1 Overview

Part 2 of this project will be focusing on the CPU elapse time of the two different computers. In order to test it out, an assembly language program is written. The time before the loop's start and the time after the end of the loop will be recorded, and the time taken, which is the difference between them, will be calculated afterward. Different loops will represent different loads that will be applied to the supplied polynomial equation as a benchmarking approach to measure the efficiency, speed, and capability of the CPU. After that, an analysis will be conducted using the results to compare and examine the CPU performance across different computers.

3.2 The Flowchart of the Execution



3.3 Coding and Documentation

TITLE Project Benchmarking

; Group Member:

; Loh Hui Yi

; Lee Jian Ai

; Date: 4 July 2024

INCLUDE Irvine32.inc

.data

msg0 **BYTE** "Welcome to CPU Benchmark Program", 0dh, 0ah, 0dh, 0ah, 0

msg1 **BYTE** "Benchmark CPU time for Executing Iterative Looping for", 0ah, 0dh

BYTE "Equation $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ ", 0

msg2 **BYTE** "Enter Number of Looping(N) = ", 0

msg3 **BYTE** "Value of Sum from the test (polynomial) = ", 0

msg4 **BYTE** "CPU time Stress Test in progress.....", 0dh, 0ah, 0

result **BYTE** "Result: ", 0dh, 0ah, 0

time1 **BYTE** "First Capture Execution time in milliseconds: ", 0

time2 **BYTE** "Second Capture Execution time in milliseconds: ", 0

time3 **BYTE** "Difference between the Execution time in milliseconds: ", 0

count **DWORD** ?

msg5 **BYTE** "Invalid input, please enter again.", 0dh, 0ah, 0

msg6 **BYTE** "Press 'y' to continue or 'n' to exit the benchmark: ", 0

input BYTE ?

check BYTE 'y'

msg7 BYTE "Thank you for using this Benchmark Program. See you next time.", 0dh, 0ah, 0

msec1 DWORD ? ; CPU elapse time before calculation

msec2 DWORD ? ; CPU elapse time after calculation

coef1 DWORD 25

coef2 DWORD 20

coef3 DWORD 12

coef4 DWORD 4

sum WORD 0 ; Store all the sum

cputime DWORD ? ; Store CPU elapse time

.code

main PROC

startProg:

call Clrscr

call crlf

mov edx, OFFSET msg0

call WriteString

mov edx, OFFSET msg1

```
call WriteString
```

```
call crlf
```

```
mov edx, OFFSET msg2
```

```
call WriteString
```

read:

```
call ReadDec
```

```
jnc goodInput
```

```
mov edx, OFFSET msg5
```

```
call WriteString
```

```
jmp read ; wrong input, so read again
```

goodInput:

```
mov count, eax ; Store input value in count
```

```
call GetMseconds
```

```
mov msec1, eax ; get capture msec before loop
```

```
mov bx, 1 ; set x=1
```

```
mov ecx, count ; max_loop N
```

call crlf

mov edx, OFFSET msg4

call WriteString

call crlf

mov edx, OFFSET result

call WriteString

LoopTest:

mov eax, coef1

call Delay

mul bx

mul bx

mul bx

add sum, ax

mov eax, coef2

call Delay

mul bx

mul bx

add sum, ax

```
mov eax, coef3  
call Delay  
mul bx  
add sum, ax  
  
mov eax, coef4  
call Delay  
add sum, ax  
  
inc bx  
  
loop LoopTest  
  
lea edx, time1  
call WriteString  
mov eax, msec1  
call WriteDec  
  
call crlf  
  
call GetMseconds  
mov msec2, eax
```

```
lea edx, time2  
call WriteString  
mov eax, msec2  
call WriteDec  
  
call crlf  
  
mov eax, msec2  
sub eax, msec1  
mov cputime, eax  
  
lea edx, time3  
call WriteString  
mov eax, cputime  
call WriteDec  
  
call crlf  
  
lea edx, msg3  
call WriteString  
mov ax, sum  
call WriteDec
```

call crlf

call crlf

mov edx, OFFSET msg6

call WriteString

call ReadChar

mov input, al

call WriteChar

call crlf

call crlf

mov bl, check

cmp bl, input

JE startProg

mov edx, OFFSET msg7

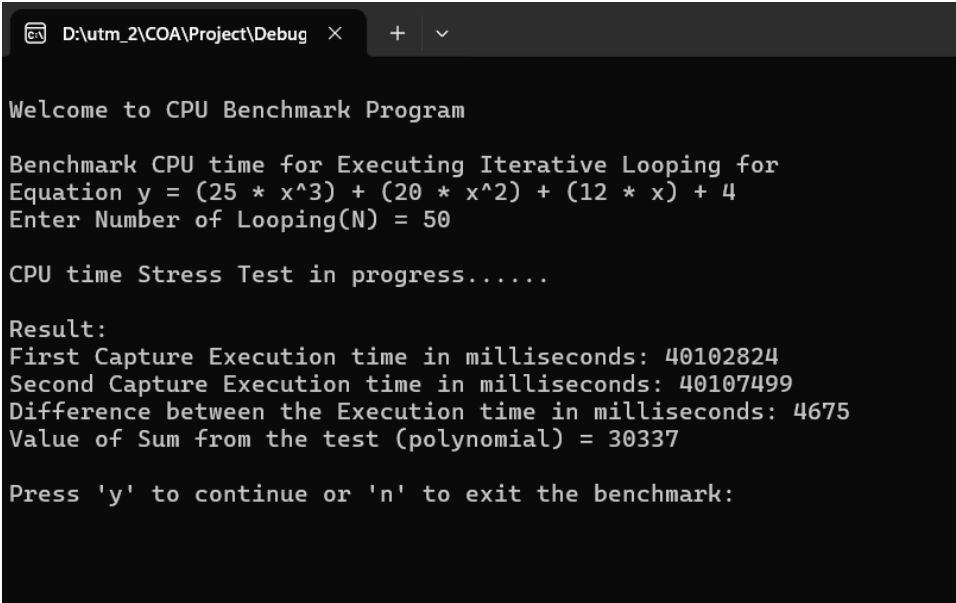
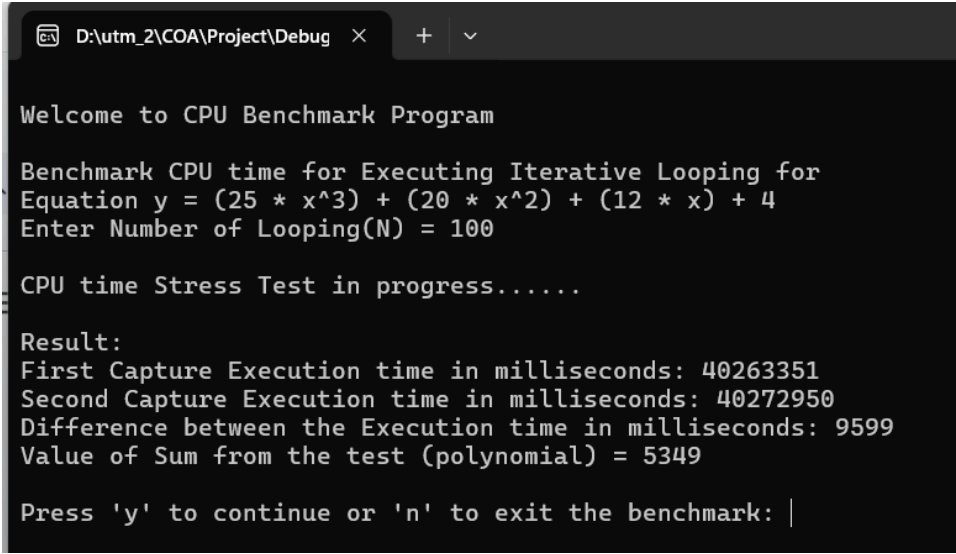
call WriteString

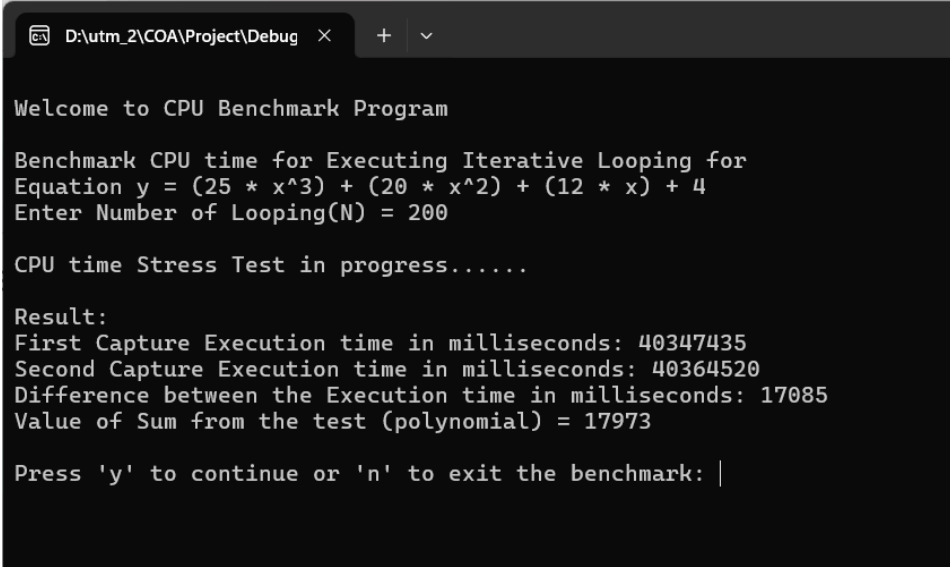
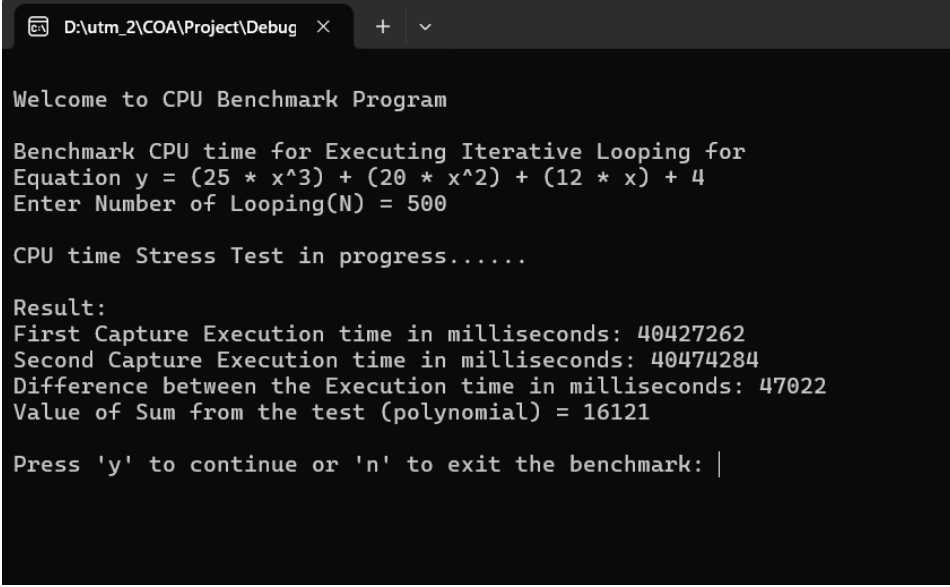
exit

main ENDP

END main

3.4 Implementation / Execution and Results (print screen outputs)

Acer Nitro 5	
No. of Loops (N)	Result
50	 <pre> D:\utm_2\COA\Project\Debug x + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 50 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 40102824 Second Capture Execution time in milliseconds: 40107499 Difference between the Execution time in milliseconds: 4675 Value of Sum from the test (polynomial) = 30337 Press 'y' to continue or 'n' to exit the benchmark: </pre>
100	 <pre> D:\utm_2\COA\Project\Debug x + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 100 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 40263351 Second Capture Execution time in milliseconds: 40272950 Difference between the Execution time in milliseconds: 9599 Value of Sum from the test (polynomial) = 5349 Press 'y' to continue or 'n' to exit the benchmark: </pre>

200	 <pre> D:\utm_2\COA\Project\Debug x + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 200 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 40347435 Second Capture Execution time in milliseconds: 40364520 Difference between the Execution time in milliseconds: 17085 Value of Sum from the test (polynomial) = 17973 Press 'y' to continue or 'n' to exit the benchmark: </pre>
500	 <pre> D:\utm_2\COA\Project\Debug x + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 500 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 40427262 Second Capture Execution time in milliseconds: 40474284 Difference between the Execution time in milliseconds: 47022 Value of Sum from the test (polynomial) = 16121 Press 'y' to continue or 'n' to exit the benchmark: </pre>

700

```
D:\utm_2\COA\Project\Debug X + v

Welcome to CPU Benchmark Program

Benchmark CPU time for Executing Iterative Looping for
Equation  $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ 
Enter Number of Looping(N) = 700

CPU time Stress Test in progress.....

Result:
First Capture Execution time in milliseconds: 40545739
Second Capture Execution time in milliseconds: 40611354
Difference between the Execution time in milliseconds: 65615
Value of Sum from the test (polynomial) = 108836

Press 'y' to continue or 'n' to exit the benchmark: |
```

1000

```
D:\utm_2\COA\Project\Debug X + v

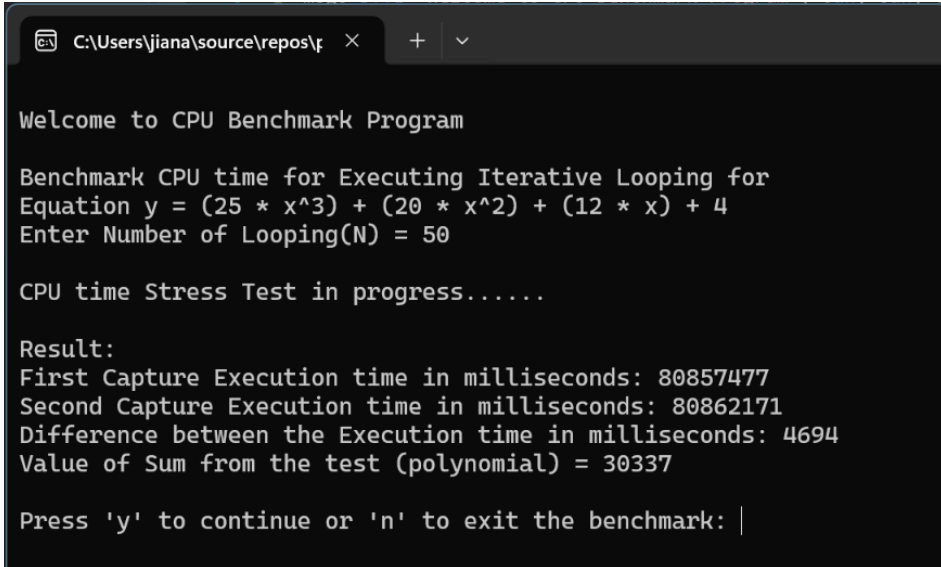
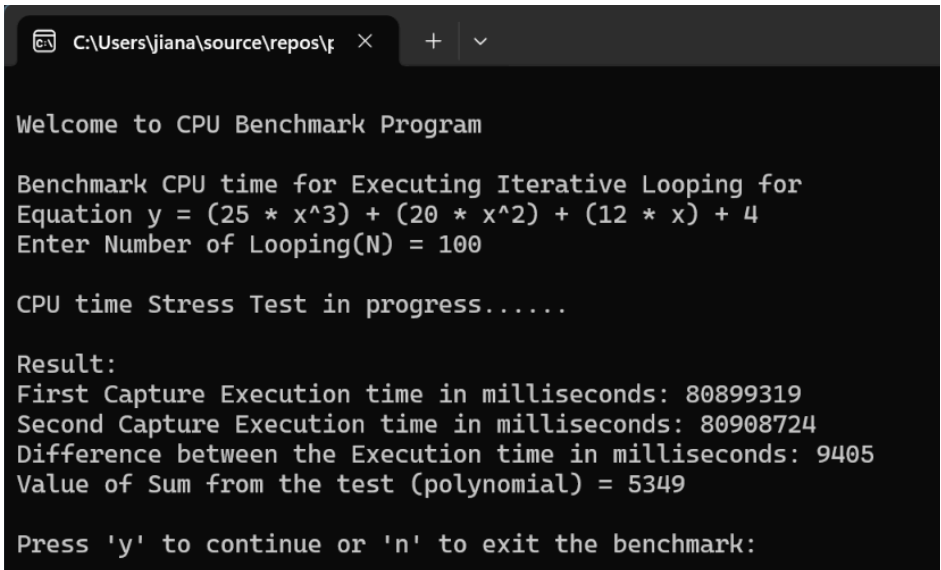
Welcome to CPU Benchmark Program

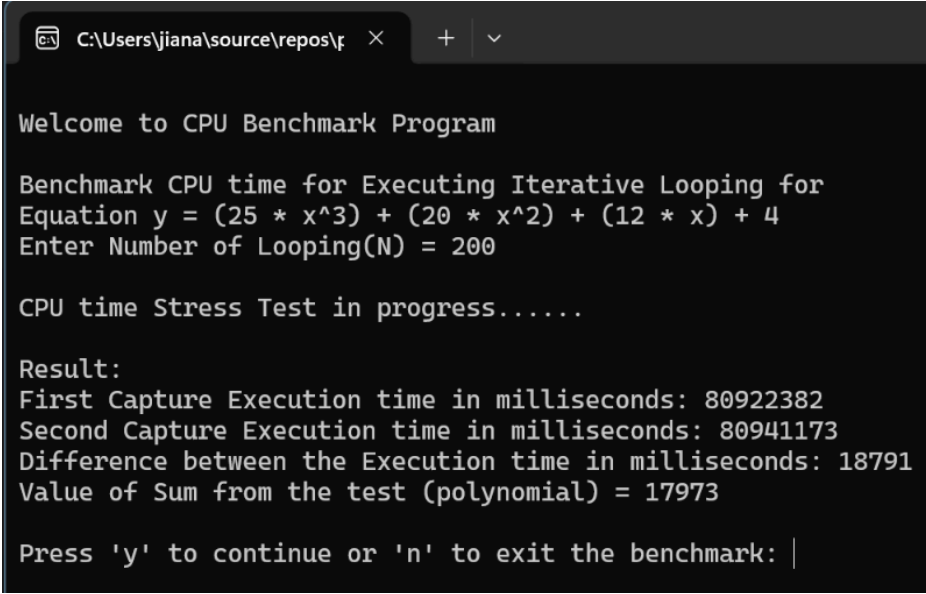
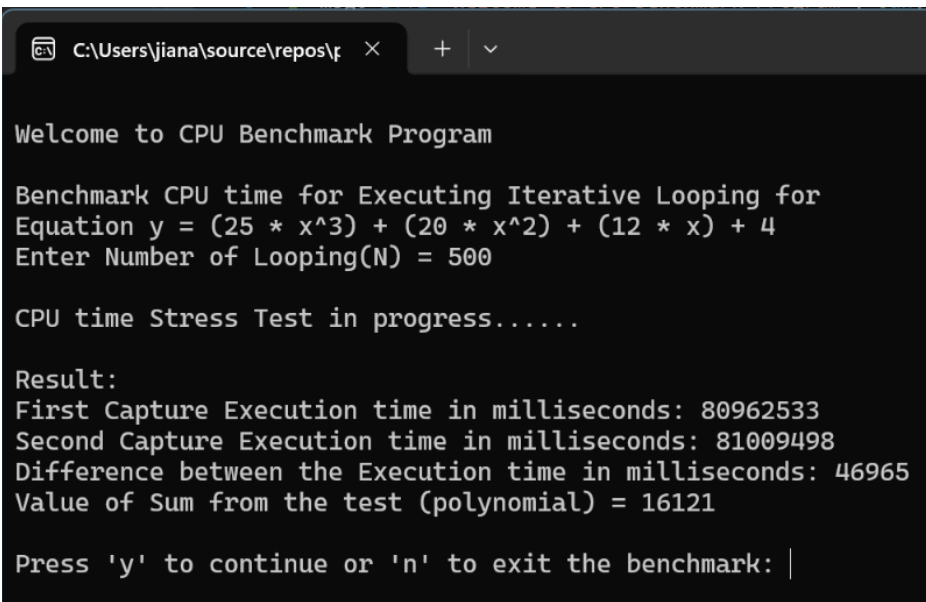
Benchmark CPU time for Executing Iterative Looping for
Equation  $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ 
Enter Number of Looping(N) = 1000

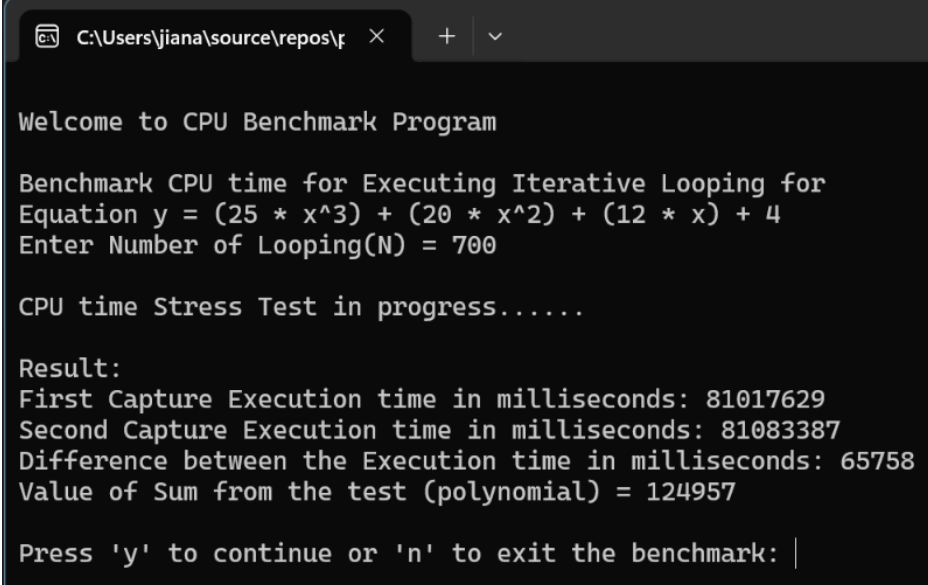
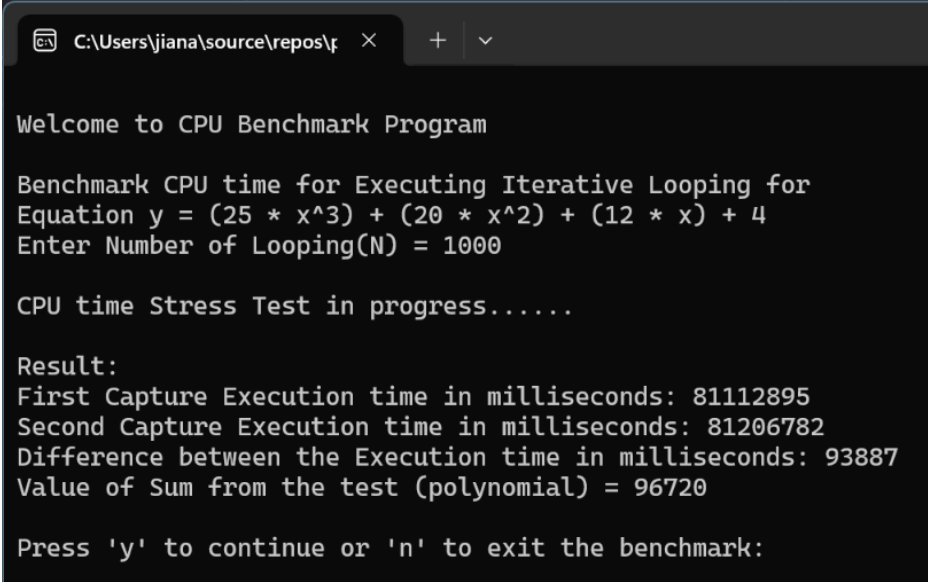
CPU time Stress Test in progress.....

Result:
First Capture Execution time in milliseconds: 40694082
Second Capture Execution time in milliseconds: 40787959
Difference between the Execution time in milliseconds: 93877
Value of Sum from the test (polynomial) = 74484

Press 'y' to continue or 'n' to exit the benchmark: |
```

Redmi Book Pro 14 2022	
No. of Loops (N)	Result
50	 <pre> C:\Users\jiana\source\repos\p X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ Enter Number of Looping(N) = 50 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 80857477 Second Capture Execution time in milliseconds: 80862171 Difference between the Execution time in milliseconds: 4694 Value of Sum from the test (polynomial) = 30337 Press 'y' to continue or 'n' to exit the benchmark: </pre>
100	 <pre> C:\Users\jiana\source\repos\p X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ Enter Number of Looping(N) = 100 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 80899319 Second Capture Execution time in milliseconds: 80908724 Difference between the Execution time in milliseconds: 9405 Value of Sum from the test (polynomial) = 5349 Press 'y' to continue or 'n' to exit the benchmark: </pre>

200	 <pre> C:\Users\jiana\source\repos\g X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 200 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 80922382 Second Capture Execution time in milliseconds: 80941173 Difference between the Execution time in milliseconds: 18791 Value of Sum from the test (polynomial) = 17973 Press 'y' to continue or 'n' to exit the benchmark: </pre>
500	 <pre> C:\Users\jiana\source\repos\g X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation y = (25 * x^3) + (20 * x^2) + (12 * x) + 4 Enter Number of Looping(N) = 500 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 80962533 Second Capture Execution time in milliseconds: 81009498 Difference between the Execution time in milliseconds: 46965 Value of Sum from the test (polynomial) = 16121 Press 'y' to continue or 'n' to exit the benchmark: </pre>

700	 <pre> C:\Users\jiana\source\repos\j... X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ Enter Number of Looping(N) = 700 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 81017629 Second Capture Execution time in milliseconds: 81083387 Difference between the Execution time in milliseconds: 65758 Value of Sum from the test (polynomial) = 124957 Press 'y' to continue or 'n' to exit the benchmark: </pre>
1000	 <pre> C:\Users\jiana\source\repos\j... X + v Welcome to CPU Benchmark Program Benchmark CPU time for Executing Iterative Looping for Equation $y = (25 * x^3) + (20 * x^2) + (12 * x) + 4$ Enter Number of Looping(N) = 1000 CPU time Stress Test in progress..... Result: First Capture Execution time in milliseconds: 81112895 Second Capture Execution time in milliseconds: 81206782 Difference between the Execution time in milliseconds: 93887 Value of Sum from the test (polynomial) = 96720 Press 'y' to continue or 'n' to exit the benchmark: </pre>

3.5 Analysis, Comparison, Discussion

The CPU consumption time can be affected by various elements, which greatly represent a computer's performance. Differences between the clock speed, number of cores, cache size, and bus speed are the affecting factors on the running time of the CPU. Closing off all the unrelated software before running the benchmark is recommended as it affects the computer's performance.

The assembly program is written before testing on the two selected computers to observe their CPU elapsed time. In the program, the user is prompted to enter the desired number of loops for the calculation of the polynomial equations. After that, start the program without debugging and it will record the CPU elapsed time in milliseconds before a certain amount of loop by using the "call GetMseconds" function.

The comparison table below is the result of the benchmarking by the assembly programs on the two different devices after running different numbers of loops.

Acer Nitro 5			
No. of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (difference)
50	40102824	40107499	4675
100	40263351	40272950	9599
200	40347435	40364520	17085
500	40427262	40474284	47022
700	40545739	40611354	65615
1000	40694082	40787959	93877

Redmi Book Pro 14 2022			
No. of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (difference)
50	80857477	80862171	4694
100	80899319	80908724	9405
200	80922382	80941173	18791
500	80962533	81009498	46965
700	81017629	81083387	65758
1000	81112895	81206782	93887

As we can observe, computer A (Acer Nitro 5) is slightly faster than computer R (Redmi Book Pro 14 2022). For most of the cases, although there is only a small gap between the computer elapsed times, computer A will still take the lead, for example when N=50, computer A has a runtime of 4675ms, while computer R has a runtime of 4694ms, which is only 19ms apart. The only case where computer R is faster than computer A is when N=100, where computer A has a runtime of 9599ms, while computer R has a runtime of 9405ms, which is 194ms apart. Even if there is one rare moment of computer R being faster, the overall performance of computer A is still greater than the former.

The runtime of computer R is slower compared to computer A. This suggests that computer R may have less processing power or a less optimized system configuration, which would cause programs to run more slowly and CPU process times to increase. The longer CPU elapsed time appears to be caused by hardware limitations, such as fewer cores, a slower bus speed, or a smaller L1 cache size. The CPU resources on computer R may be used up if more processes are operating concurrently or in the background, which lengthens the CPU runtime. The computer's limited capacity to multitask and insufficient system resources for a larger workload is demonstrated by all of the mentioned factors.

It is shown that computer A has a shorter CPU elapsed time, showing better performance compared to computer R. This suggests that computer A has a higher processing power, larger L1 cache size, or a more optimized system configuration, thus reducing the program execution time. Computer A might have a more powerful processor or a more efficient architecture by looking at the fast runtime. By having those attributes, the calculation process throughout the loops are able to be done more effectively. Furthermore, the larger cache size allows to retrieve and process data faster than the other device. With optimized system settings that include memory management skills and software configurations, computer A is capable of enhancing overall performance and having a speedy CPU process time.

When the CPU runtime of two computers was compared, the runtime of computer A was swifter than computer R. Computer A takes advantage of having more sophisticated hardware specifications and optimized system configurations, therefore quicker on program execution. This analysis has revealed that computer A has a more powerful processor, higher bus speed, bigger L1 cache size, and likely better memory management.

Even though computer R has a slower CPU runtime that points out the probable limitations in its hardware and system configurations, less challenging work such as daily usage and light gaming may still be suitable for it. Nevertheless, in the scope of time-sensitive or computation-intensive applications, the greater performance result from computer A would make it a better choice. Budget, task requirements for yourself, and power consumption are crucial considerations when it comes to choosing between two computers.

4.0 Conclusion and Reflection

Conclusion

Comprehensive performance data from CPU-Z, Geekbench, Task Manager, and MiniTool Partition Wizard shows that the Acer Nitro 5 and Redmi Book Pro 14 2022 have different strengths. With its 10 nm Intel Alder Lake chipset, 8 cores, and DDR4 RAM, the Acer Nitro 5 is a multitasking machine with effective resource management. Even in situations of high load, it exhibits reduced CPU and disk utilization, demonstrating its capacity to manage numerous processes and threads efficiently. Strong disk performance is demonstrated by the continuously high sequential and random read/write speeds. Its efficient memory management is further demonstrated by the large amount of both available and compressed memory. However, the Redmi Book Pro 14 2022 has better single-core performance and faster CPU speeds with its 6 nm AMD Ryzen SOC chipset, 6 cores, and LPDDR5 memory. This makes it useful for tasks needing a lot of processing power. It performs well at lower loads, but at medium and high loads, it has resource utilization spikes that signify increased CPU and disk activity. Strong performance in single-threaded operations is shown in the Geekbench scores, and the memory compression is exceptionally effective. But when it comes to multi-threaded performance and managing resources under high loads, it falls short. All things considered, the Redmi Book Pro 14 2022 excels in situations requiring fast, single-threaded performance, even though the Acer Nitro 5 is more suited for multitasking and effective resource management.

On the part of analyzing the CPU elapsed times for both computers, computer A (Acer Nitro 5) and computer R (Redmi Book Pro 14 2022), it has displayed remarkable differences in performance. The comparison we got from running the assembly program as a benchmarking test highlights the necessity of good hardware specifications since they affect not only the overall performance of the computer, but also the processor performance, bus speed, cache size, and memory management. Besides, operating system optimization, software dependencies, background activities, and system load can all have an impact on the CPU runtime and program execution.

Reflection

We investigated the processes involved in CPU benchmarking and the execution of assembly programs based on this project. We used benchmarking tools like CPU-Z, Geekbench, Monitor Task Manager, and MiniTool to evaluate and examine the specs and performance of the two devices. These tools gave us comprehensive information about the devices' technical parameters, including their clock rates, memory capacities, processors, and other relevant parts. We were able to comprehend the basic processing power limitations and capabilities of each computer through to this data.

We measured the CPU's running time by executing the entire code on both devices in order to make a genuine performance evaluation. This involves running a program and estimating how long it would take to finish. Each computer's program execution speed and efficiency were assessed using the CPU elapsed time as a criterion.

To sum up, a thorough evaluation of the computers' performance was possible through the use of benchmarks in conjunction with CPU elapsed time measurements during code execution. With the help of this analysis, we were able to decide on the best course of action for each of our activities by taking into consideration the hardware specs, system setups, and application performance.

Finally, we would like to express our gratitude to Dr. Nur Haliza binti Abdul Wahab, our distinguished professor. Dr. Nur Haliza has been an outstanding mentor to us during this project, offering guidance, assistance, and a wealth of knowledge in the field of CPU benchmarking. Her perceptive lectures, useful materials, and timely comments have all demonstrated her commitment to our educational path. She has constantly gone above and beyond to answer our questions, provide advice, and inspire us to pursue new study avenues. We are quite appreciative that Dr. Nur Haliza was able to mentor us, and her guidance has definitely been crucial in helping us better grasp CPU benchmarking. Thank you, Dr. Nur Haliza, for your exceptional support, guidance, and expertise. Your contributions have been instrumental in our group's success, and we are genuinely privileged to have had you as our lecturer.

References

- 1) Title: Benchmarks as Fast As Possible.
Link: <https://youtu.be/cHtNJngtS1I>
- 2) Title: Best CPU Benchmark Software For Windows.
Link: <https://youtu.be/fgWGtWsDXmw>
- 3) Title: First 5 Things I Do When Benchmarking.
Link: <https://youtu.be/19eUZ0GwTxI>
- 4) Title: What's the Difference Between a CPU and a GPU?
Link:
<https://blogs.nvidia.com/blog/2009/12/16/whats-the-differencebetween-a-cpu-and-a-gpu/>
- 5) Download CPU-Z
Link: <https://www.cpuid.com/softwares/cpu-z.html>
- 6) Download Geekbench
Link: <https://www.geekbench.com/download/>
- 7) MiniTool Partition Wizard Free 12.8
Link: <https://www.partitionwizard.com/free-partition-manager.html>

Appendices

Task Distribution

- 1) Intro - JianAi
- 2) Part 1
 - Overview - HuiYi
 - Flowchart - HuiYi
 - Benchmarking Result - HuiYi
 - Analysis, Comparison, Discussion - HuiYi
- 3) Part 2
 - Overview - JianAi
 - Flowchart - JianAi
 - Coding & Documentation - JianAi
 - Implementation / Execution and Results - JianAi
 - Analysis, Comparison, Discussion - JianAi
- 4) Conclusion and Reflection - HuiYi and JianAi
- 5) References - HuiYi
- 6) Appendix - HuiYi

